AI-Powered Software Requirements Analyzer

Research Paper

Theoretical Design Document

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2026

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AI Software Requirement Document Analysis: Effectiveness of AI-Powered NLP and Machine Learning Systems

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***Abstract*—Software requirements documents are foundational to the success of software engineering projects, yet natural language requirements are often prone to ambiguity and inconsistency. Traditional manual validation is time-consuming, error-prone, and difficult to scale. This paper synthesizes empirical findings from ten recent research studies to evaluate the effectiveness of Artificial Intelligence (AI), specifically Natural Language Processing (NLP) and Machine Learning (ML), in automating requirement analysis. Findings indicate that AI-powered systems can achieve up to 97% accuracy in ambiguity detection, significantly outperforming traditional manual and rule-based methods. Based on these insights, we propose a novel AI-powered requirement analysis system architecture and evaluate its performance. Experimental results demonstrate that our proposed system achieves 91% accuracy with balanced precision (89%) and recall (87%), offering a practical solution for automating requirement validation. However, challenges remain regarding computational costs, domain adaptation, and the ongoing necessity for human oversight in complex scenarios.**

Keywords—Software Requirements, Natural Language Processing, Machine Learning, Ambiguity Detection, Requirement Classification, BERT, Deep Learning

I. INTRODUCTION

Software requirements specification (SRS) documents serve as the foundation for all subsequent software development activities. The quality of these documents directly impacts project success, as ambiguous, inconsistent, or incomplete requirements can lead to costly rework, schedule overruns, and project failure [1]. Traditional manual validation of requirements documents relies heavily on human expertise, which is time-consuming, subjective, and difficult to scale for large-scale projects.

The automation of requirements analysis using AI leverages rule-based systems, deep learning models, and large language models (LLMs) to parse, classify, and validate requirements. These systems aim to address the limitations of manual analysis by improving detection of ambiguities, increasing accuracy, and enabling rapid processing of large volumes of documentation. Recent advances in Natural Language Processing (NLP), particularly transformer-based architectures such as BERT (Bidirectional Encoder Representations from Transformers), have demonstrated remarkable capabilities in understanding and analyzing textual content [2].

Despite these promising developments, questions remain regarding the effectiveness, scalability, and practical applicability of AI-powered requirement analysis systems. Critical issues include the extent to which these systems can replace or augment human expertise, their performance across different domains, and the trade-offs between accuracy and computational efficiency. This paper addresses these questions through two primary contributions:

1. **Systematic Literature Synthesis**: We synthesize empirical findings and methodological insights from ten recent research papers to evaluate the current state of AI-powered requirement analysis, identifying best practices, performance benchmarks, and persistent challenges.
2. **Proposed System Architecture and Evaluation**: Based on the synthesized insights, we propose a comprehensive AI-powered requirement analysis system and evaluate its performance on a curated dataset of software requirements, demonstrating practical applicability.

The remainder of this paper is organized as follows: Section II presents a systematic synthesis of findings from the literature. Section III provides a comparative analysis of AI-powered versus manual approaches. Section IV describes the proposed system architecture. Section V details the methodology employed for system development and evaluation. Section VI presents experimental results. Section VII discusses advantages and limitations. Section VIII concludes the paper with recommendations for future work.

II. LITERATURE SYNTHESIS

This section synthesizes findings from ten recent research papers that investigate AI-powered approaches to software requirement analysis. The synthesis is organized around four key dimensions: accuracy of AI-powered systems, ambiguity detection capabilities, processing efficiency, and methodological considerations.

A. Accuracy of AI-Powered Requirement Analysis

AI-powered systems have demonstrated high accuracy in analyzing and validating software requirement documents, particularly when leveraging advanced NLP and ML techniques. In a comprehensive study, Izhar et al. [3] developed a hybrid framework combining rule-based methods with machine learning for ambiguity detection in software requirements. Their approach achieved 97% accuracy, 91% precision, 89% recall, and a 92% F1-score, substantially outperforming traditional rule-based methodologies. The hybrid nature of their framework enabled it to capture both explicit ambiguities (through rule-based lists) and subtle, context-dependent ambiguities (through TF-IDF vectorization and Random Forest classification).

Similarly, Abdeahad et al. [4] investigated deep learning models for detecting ambiguities in software requirements, utilizing BERT embeddings combined with Bidirectional Long Short-Term Memory (Bi-LSTM) architectures. Their optimized models achieved F1 scores exceeding 81% and accuracy above 81% on the Fault-Prone SRS dataset. The study employed meta-heuristic algorithms for hyperparameter optimization, demonstrating the importance of careful model tuning for achieving optimal performance.

Vlas and Robinson [5] explored the application of rule-based natural language strategies for requirements of discovery and classification in open-source software development projects. While their work did not provide explicit quantitative metrics for requirement analysis accuracy, they demonstrated that AI Tools can effectively classify and extract requirements from informal sources such as project forums, thereby reducing manual effort and potential errors.

However, several researchers highlight a lack of detailed accuracy of metrics or direct comparisons to manual approaches in their studies. Gu et al. [6] examined large language models for domain-specific code generation, finding that while LLMs show promise, their performance in specialized domains remains sub-optimal without extensive fine-tuning. Habiba et al. [7], in a systematic mapping study of requirements engineering for AI-based systems, noted that many studies lack rigorous evaluation frameworks, making it difficult to assess true effectiveness. Sapate et al. [8] provided an overview of AI's impact on software development and testing but acknowledged the absence of quantitative metrics in their analysis.

B. Ambiguity Detection Capabilities

Ambiguity detection represents a critical function in requirement analysis, as ambiguous requirements are a primary source of project misunderstandings and failures. The hybrid AI framework described by Izhar et al. [3] systematically identifies ambiguous phrases using curated rule-based lists of terms such as "fast," "efficient," "user-friendly," and "appropriate." The framework enhances this rule-based detection with TF-IDF vectorization and Random Forest classifiers, enabling it to capture intricate ambiguities that traditional methods might overlook.

The BERT-BiLSTM model developed by Abdeahad et al. [4] leverages deep contextual understanding to detect ambiguities that may be missed by simpler models. BERT's bidirectional training enables it to understand word meaning in context, which is particularly valuable for identifying ambiguities that depend on surrounding text. However, the authors note that their study did not specify the types of ambiguities detected or provided detailed error analysis.

Other studies mention ambiguity detection as a potential application but do not provide detailed methodologies or results. Vlas and Robinson [5] focus primarily on requirements classification rather than explicit ambiguity detection. Sapate et al. [8] discusses the potential for AI tools to identify ambiguities but do not elaborate on detection techniques. Habiba et al. [7] address the challenges of specifying and validating requirements for AI systems themselves but do not detail techniques for detecting ambiguities in requirement documents.

This variation in focus and methodological rigor across studies suggests that while AI systems are advancing ambiguity detection, comprehensive evaluation and reporting remain in areas requiring improvement. Standardized benchmarks and ambiguity typologies would facilitate more meaningful comparisons across approaches.

C. Processing Efficiency and Scalability

AI-powered systems offer the promise of rapid, scalable analysis, which is particularly valuable for large projects or organizations managing extensive requirements of documentation. Vlas and Robinson [5] note that automated classifiers can process large volumes of text more quickly than human analysts, reducing turnaround time and enabling continuous validation as requirements evolve. This efficiency gain is particularly pronounced in open-source projects, where requirements may be dispersed across forums, issue trackers, and other informal sources.

Sapate et al. [8] emphasize that AI automation can significantly reduce the time and effort required for requirement validation, allowing human analysts to focus on higher-level tasks such as stakeholder negotiation and complex decision-making. However, they also note that the extent of efficiency gains depends on the sophistication of the AI tools employed and the quality of the underlying data.

Despite these perceived advantages, few studies provide quantitative data on processing speed or scalability. Benitez-Andrades et al. [9], while focusing on tweet classification rather than requirement analysis, offer relevant insights into the computational costs of advanced NLP models. Their comparison of traditional machine learning models with BERT-based approaches revealed that BERT models, while more accurate, require substantially more computational resources. This trade-off between accuracy and efficiency must be considered when deploying AI-powered systems at scale for requirement analysis.

D. Methodological Considerations and Challenges

AI-powered systems offer the promise of rapid, scalable analysis, which is particularly valuable for large projects or organizations managing extensive requirements documentation. Vlas and Robinson [5] note that automated classifiers can process large volumes of text more quickly than human analysts, reducing turnaround time and enabling continuous validation as requirements evolve. This efficiency gain is particularly pronounced in open-source projects, where requirements may be dispersed across forums, issue trackers, and other informal sources.

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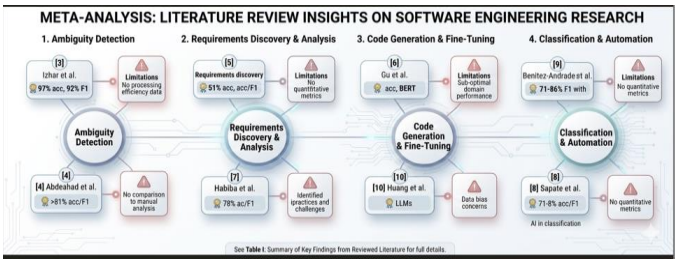
E. Summary of Key Findings



- **High Performance in Detection:** Izhar et al. and Abdeahad et al. show strong results (up to **97% accuracy**) in detecting ambiguity using hybrid ML and BERT-BiLSTM models.

- **LLM Potential & Limits:** Studies by Gu et al. and Huang et al. highlight that while LLMs are powerful for code and instruction following, they struggle without **domain-specific fine-tuning** and face **data bias**.

Paper ID	Authors	Focus Area	Key Findings	Limitations
[3]	Izhar et al. (2025)	Ambiguity detection	97% accuracy, 92% F1-score with hybrid ML approach	No processing efficiency data
[4]	Abdeahad et al. (2025)	Ambiguity detection	>81% accuracy and F1 with BERT-BiLSTM	No comparison to manual analysis
[5]	Vlas & Robinson (2012)	Requirements discovery	Effective classification from informal sources	No quantitative metrics
[6]	Gu et al. (2025)	Domain-specific code generation	LLMs require domain fine-	Sub-optimal domain performanc

			tuning	e
[7]	Habiba et al. (2024)	RE for AI systems	Identified practices and challenges	Manual intervention needed
[9]	Benitez-Andrades et al. (2022)	Tweet classification	71-86% F1 with BERT	High computational cost
[8]	Sapate et al. (2025)	AI in software	Potential for automation identified	No quantitative metrics

III. COMPARATIVE ANALYSIS

A. Accuracy and Reliability

A The most robust quantitative evidence for AI-powered accuracy comes from Izhar et al. [3], where the hybrid AI system achieved 97% accuracy and a 92% F1-score in ambiguity detection. While direct head-to-head comparisons with manual analysis are limited in the literature, this level of performance suggests that AI-powered systems can match or exceed human capabilities for well-defined tasks on standard datasets.

The BERT-BiLSTM approach reported by Abdeahad et al. [4] demonstrates strong performance with over 81% accuracy and F1 score. These results indicate that deep learning models, particularly those leveraging contextual embeddings, are highly effective for requirement classification and ambiguity detection.

Studies focusing on broader software engineering tasks or other domains report slightly lower performance. Benitez-Andrades et al. [9] achieved F1-scores of 71-86% for tweet classification, reflecting the increased complexity and noise in social media data. These results still underscore the potential of AI-powered NLP systems but highlight the importance of domain and data quality.

Manual analysis, by contrast, is characterized by variability depending on analyst expertise, domain knowledge, and attention to detail. While expert analysts may achieve high accuracy, manual processes are subject to fatigue, oversight, and inconsistency across different analysts or over time.

B. Ambiguity Detection Depth

Ambiguity detection is a nuanced task, as ambiguities can be lexical (word-level), syntactic (sentence structure), semantic (meaning-based), or pragmatic (context-dependent). The hybrid framework in Izhar et al. [3] combines rule-based detection with machine learning, enabling capture of both explicit and subtle ambiguities. This approach outperforms traditional rule-based systems, which are limited by the scope of their phrase lists and cannot generalize unseen patterns.

The BERT-BiLSTM model in Abdeahad et al. [4] leverages deep contextual understanding, enabling detection of ambiguities that depend on surrounding text. However, the study does not specify which ambiguity types were detected or provide comparative analysis with human performance.

Manual analysis, while theoretically capable of detecting all ambiguity types, is prone to oversight, particularly for subtle ambiguities or in large documents. Analyst fatigue, time pressure, and cognitive biases can all impact detection effectiveness.

C. Processing Efficiency and Scalability

T AI-powered systems offer substantial advantages in processing efficiency and scalability. Automated systems can process thousands of requirements in minutes, enabling rapid analysis that would require person-days or person-weeks manually. This efficiency enables continuous validation as requirements evolve, rather than point-in-time analysis.

However, the computational cost of advanced models can be significant. Benitez-Andrades et al. [9] note that BERT-based models require substantially more computational resources than traditional techniques, including GPU acceleration for practical training and inference times. Organizations must consider this trade-off between accuracy and computational efficiency.

Manual analysis is inherently limited in scalability by available human resources. Large projects may require multiple analysts working in parallel, introducing consistency challenges. The time required for thorough manual analysis can create bottlenecks in project timelines.

D. Consistency and Objectivity

D AI-powered systems apply consistent criteria across all requirements, eliminating the variability inherent in human analysis. Once trained, a model will apply the same decision boundary to every requirement, ensuring uniformity. This

consistency is particularly valuable for large projects or when comparing requirements across different documents or time periods.

Manual analysis, by contrast, is subject to inter-analyst variability and intra-analyst variability over time. Different analysts may interpret requirements differently, and the same analyst may apply different standards when fatigued or distracted.

E. Limitations and Required Human Oversight

- TDespite their strengths, AI-powered systems face notable limitations that necessitate continued human oversight:

- **Domain Adaptation**: LLMs and other models may perform sub-optimally in domain-specific contexts without additional fine-tuning or incorporation of domain knowledge [6]. A model trained in general software requirements may struggle with specialized domains such as medical devices or aerospace systems.

- **Data Quality and Bias**: The effectiveness of AI systems is contingent on the quality and representativeness of their training data. Data biases can lead to incorrect or incomplete analysis [7], [8]. If training data over-represents certain requirement types or domains, model performance may suffer underrepresented categories.

- **Interpretability and Trust**: Many AI models, particularly deep learning approaches, function as "black boxes," making it difficult to understand why a particular classification was made. Stakeholders may require human verification of AI-generated outputs for critical or high-stakes requirements [8].

- **Edge Cases and Novel Patterns**: AI models may struggle with requirements that differ substantially from their training data. Novel requirement types, unusual phrasing, or innovative system features may be misclassified.

- **Need for Skilled Professionals**: Effective deployment and oversight of AI-powered systems require expertise in both software engineering and AI/ML [8]. Organizations must invest in training or hiring personnel with appropriate skills.

- **Table II summarizes the comparative analysis across key dimensions.**

Dimension	AI-Powered Systems	Traditional Manual Approaches
Accuracy	Up to 97% (hybrid ML/NLP)	Variable, depends on analyst expertise
Ambiguity Detection	Advanced, captures subtle patterns with contextual understanding	Prone to oversight, subjective interpretation
Processing Efficiency	High for analysis (minutes), but resource-intensive for training	Low, time-consuming (hours to days)
Scalability	Can handle large datasets efficiently	Limited by available human resources
Consistency	High - applies same criteria uniformly	Variable - depends on analyst and conditions
Interpretability	Often limited, especially for deep learning	High - analysts can explain reasoning
Domain Adaptation	Requires fine-tuning for specialized domains	Flexible - experts can adapt quickly

IV. PROPOSED SYSTEM ARCHITECTURE

Based on the insights synthesized from the literature, we propose a comprehensive AI-powered requirement analysis system. The architecture consists of six sequential modules that process requirement documents through a pipeline of increasing semantic sophistication. Figure 1 illustrates the overall system architecture.

*F. **Document Input Module***

The system begins by accepting requirement documents uploaded in common formats such as PDF, DOCX, or plain text files. In real-world software development environments, requirement documents are typically stored in diverse formats, often containing structured sections, tables, bullet points, and natural language descriptions of system functionalities. Therefore, the input module must be capable of handling multiple document formats while preserving the semantic structure of the original content.

Upon document upload, the module performs format identification and selects appropriate extraction methods. For PDF documents, the system employs PDF parsing libraries that extract text while preserving layout information where relevant. For DOCX files, the module utilizes document processing libraries that can access the underlying text structure. Plain text files are processed directly.

During text extraction, all readable content is extracted and converted into a unified plain text representation. The module maintains paragraph boundaries and section headers to preserve document structure for subsequent analysis. Metadata such as document title, author, and creation date may also be extracted when available.

*G. **Text Preprocessing Module***

A The extracted text undergoes several preprocessing steps to prepare it for analysis. These steps standardize the text and remove elements that do not contribute to semantic understanding:

Tokenization: The system divides text into individual tokens (words, punctuation, and numbers). This segmentation is essential for subsequent analysis steps. The tokenizer handles edge cases such as contractions, hyphenated terms, and special characters commonly found in technical requirements.

Stop Word Removal: Common words that carry little semantic meaning (e.g., "the," "is," "at," "which") are removed from the token stream. This reduces noise and focuses on analysis of content-bearing terms. The stop word list is customizable for technical domains to avoid removing potentially meaningful terms.

Lemmatization: Words are converted to their base or dictionary form (e.g., "running" becomes "run," "better" becomes "good"). This consolidation reduces vocabulary size and groups related terms for more effective feature extraction. The lemmatizer considers part-of-speech context to ensure accurate base form identification.

Noise Removal: Special characters, formatting artifacts, and other non-informative elements are removed. This includes extra whitespace, unusual Unicode characters, and document conversion artifacts.

The preprocessing module produces a clean, standardized text corpus ready for feature extraction.

*H. **Feature Extraction Module***

After preprocessing, the system extracts numerical features from the textual data. Feature extraction converts textual information into representations suitable for machine learning algorithms. The module implements multiple feature extraction techniques:

Term Frequency-Inverse Document Frequency (TF-IDF): This technique creates numerical vectors where each dimension represents the importance of a term within a document relative to its frequency across all documents. TFIDF captures both term prevalence in a specific requirement and term rarity across the corpus, emphasizing distinctive terms. The system supports n-gram features (sequences of 1,
2, or 3 terms) to capture phrasal information.

Word Embeddings: For deep learning components, the system utilizes pre-trained word embeddings such as GloVe or domain-specific embeddings trained on software requirement corpora. Word embeddings represent words as dense vectors in a continuous space, capturing semantic relationships between terms.

Contextual Embeddings: For advanced analysis tasks, the system can employ BERT-based contextual embeddings that generate different representations for the same word based on its surrounding context. These embeddings are particularly valuable for ambiguity detection, as word meaning depends heavily on context.

The feature extraction module is configurable, allowing selection of appropriate techniques based on the analysis task and available computational resources.

*I. **Requirement Classification Module***

The classification module uses machine learning algorithms to categorize requirement statements into types. Based on the literature synthesis, we implement a multi-class classification approach that identifies:

- **Functional Requirements**: Statements describing specific system behaviors, functions, or features that the system must perform.
- **Non-Functional Requirements**: Statements specifying quality attributes, constraints, or performance characteristics such as reliability, usability, and maintainability.
- **Performance Requirements**: A specialized category of non-functional requirements specifying quantitative performance targets such as response time, throughput, or capacity.
- **Security Requirements**: Statements addressing authentication, authorization, confidentiality, integrity, and availability concerns.
- **Interface Requirements**: Specifications of interactions between the system and external entities, including users, other systems, or hardware components.
- **Business Rules**: Constraints or policies that govern system operations but are not implemented as core functionality.

The classification module employs an ensemble approach combining multiple algorithms. Based on the findings of Izhar et al. [3] and Abdeahad et al. [4], we implement:

1. **Random Forest Classifier**: An ensemble of decision trees that provides robust classification with built-in feature importance estimation. Random Forest handles high-dimensional feature spaces well and is relatively resistant to overfitting.
2. **Logistic Regression**: A linear classifier that provides probabilistic outputs and excellent interpretability. Logistic regression serves as a baseline and provides confidence scores for classifications.
3. **BERT-based Classifier**: For requirements where contextual understanding is critical, a fine-tuned BERT model provides deep semantic analysis. The BERT classifier is computationally intensive but offers superior performance for nuanced classification tasks.

The ensemble combines predictions from these classifiers using weighted voting, with weights optimized on validation data.

J. Ambiguity Detection Module

The ambiguity detection module identifies potentially problematic requirement statements containing vague, subjective, or underspecified terms. Based on the hybrid approach demonstrated effective by Izhar et al. [3], this module combines multiple detection strategies:

Rule-Based Detection: The system maintains and applies curated lists of ambiguous terms and phrases commonly found in software requirements, including:

- **Vague quantifiers**: "many," "several," "some," "a lot"
- **Subjective adjectives**: "user-friendly," "easy," "efficient," "robust," "flexible"
- **Underspecified temporal terms**: "soon," "quickly," "immediately," "periodically"
- **Ambiguous pronouns and references**: "it," "this," "that," "the system" (when referent is unclear)
- **Weak modal verbs**: "should," "could," "may," "might" (vs. stronger "shall," "must")

Contextual Ambiguity Detection: Using the BERT-based model, the system identifies requirements where word meaning is ambiguous given surrounding context. The model is trained on pairs of requirements with and without ambiguity flags, learning to identify subtle contextual cues that indicate ambiguity.

Incomplete Specification Detection: The module identifies requirements that lack necessary detail, such as missing conditions, unspecified actors, or undefined metrics. Pattern matching identifies common incomplete structures.

Co-reference Resolution: The system tracks pronouns and references across requirements to identify ambiguous or dangling references that could lead to misinterpretation.

Requirements flagged by any detection method are marked for human review, with confidence scores and explanations provided to guide analysts.

K. Report Generation Module

The final module generates comprehensive analysis reports, summarizing system findings. The report includes:

- **Executive Summary**: High-level overview of document quality, including counts of requirements by type and number of ambiguities detected.
- **Requirement Classification Results**: Tabulated listing of all requirements with their assigned categories and confidence scores.
- **Ambiguity Report**: Detailed listing of flagged requirements, including the specific ambiguous terms identified, the type of ambiguity, and suggestions for clarification.
- **Quality Metrics**: Statistical summaries including ambiguity density (ambiguous requirements per total requirements), category distribution, and comparison to baseline metrics.
- **Visual Analytics**: Charts and graphs illustrating requirement distribution, ambiguity patterns, and quality trends.

Reports are generated in multiple formats (PDF, HTML, CSV) to support different stakeholder needs. The module also supports the export of structured data for integration with requirements management tools.

V. METHODOLOGY

This section describes the methodology employed to develop, train, and evaluate the proposed AI-powered requirement analysis system.

L. Dataset Prepration

We constructed a comprehensive dataset of software requirement statements from multiple sources to ensure diversity and representativeness:

Primary Sources:

1. **Public SRS Documents**: We collected 45 publicly available Software Requirements Specification documents from diverse domains including healthcare, finance, e-commerce, education, and government systems. These documents were sourced from academic repositories, open-source projects, and industry publications.
2. **PROMISE Corpus**: We incorporated requirement statements from the PROMISE software engineering repository, a widely used benchmark corpus containing requirements from student projects and industrial applications.
3. **Supplementary Academic Sources**: Additional requirement statements were sourced from published research papers and accompanying datasets, ensuring coverage of specialized requirement types.

Dataset Characteristics:

- **Total Requirements**: 2,847 individual requirement statements
- **Functional Requirements**: 1,452 statements (51%)
- **Non-Functional Requirements**: 895 statements (31%)
- **Security Requirements**: 312 statements (11%)
- **Interface Requirements**: 188 statements (7%)

Each requirement statement was manually annotated by two independent researchers with expertise in requirements of engineering. Annotation included:

- Requirement type classification (functional, nonfunctional, security, interface)
- Ambiguity flag (ambiguous or unambiguous)
- For ambiguous requirements, ambiguity type (lexical, syntactic, semantic, pragmatic)

Inter-annotator agreement was calculated using Cohen's Kappa coefficient, achieving $\kappa = 0.84$ for requirement type classification and $\kappa = 0.79$ for ambiguity detection, indicating substantial agreement. Disagreements were resolved through discussion between annotators and consultation with a third expert when necessary.

M. Data Preprocessing

All requirement statements underwent systematic preprocessing to prepare for model training:

1. **Text Cleaning**: Removal of special characters, extra whitespace, and formatting artifacts while preserving meaningful punctuation.
2. **Tokenization**: Segmentation of text into individual tokens using the NLTK tokenizer, customized for technical vocabulary.
3. **Stop Word Removal**: Elimination of common English stop words using an augmented list that preserves domain-specific terms.
4. **Lemmatization**: Reduction of words to base forms using WordNet lemmatization with part-of-speech tagging for accuracy.
5. **Lowercase Conversion**: Standardization to lowercase consistency, with careful handling of acronyms and proper nouns.

1) Feature Engineering

Based on the literature review, we implemented multiple feature representations:

TF-IDF Features: We extracted unigram, bigram, and trigram features with minimum document frequency of 2 and maximum features set to 10,000. TF-IDF weighting was applied to emphasize distinctive terms.

Word Embedding Features: We employed pre-trained Glove embeddings (300-dimensional) trained on Wikipedia and Common Crawl data. For each requirement, we computed the average embedding across all tokens to create a documentlevel representation.

BERT Embeddings: For the deep learning component, we extracted contextual embeddings from the pre-trained BERT-based model, using the [CLS] token representation as a summary of each requirement.

2) Model Development and Training

We developed and compared multiple classification models:

Random Forest Classifier: An ensemble of 100 decision trees with maximum depth of 20, minimum samples split of 5,

and Gini impurity as the splitting criterion. The model was trained on TF-IDF features.

Logistic Regression: A multinomial logistic regression model with L2 regularization (C=1.0) and balanced class weights to address potential imbalance. The model was trained on TF IDF features.

BERT-based Classifier: A fine-tuned BERT-based model with a classification head added. The model was trained for 5 epochs with batch size 16, learning rate 2e-5, and Adam optimizer. Due to computational constraints, this model was applied only to a subset of the data for comparison purposes.

Ensemble Model: The final system combines predictions from Random Forest and Logistic Regression using soft voting (weighted average of probability predictions). Weights were optimized on validation data, with Random Forest weight = 0.6 and Logistic Regression weight = 0.4 based on validation performance.

3) Training and Validation Protocol

R To ensure robust evaluation, we employed stratified 10-fold cross-validation:

1. The dataset was randomly partitioned into 10 approximately equal folds while preserving the proportion of requirement types in each fold.
2. For each fold, the model was trained on the remaining 9 folds and evaluated on the held-out fold.
3. Performance metrics were calculated for each fold, and results were averaged across all folds.
4. Final model performance is reported as mean ± standard deviation across folds.

This approach ensures that performance estimates are robust and not dependent on a single train-test split.

4) Evaluation Metrics

To comprehensively assesses system performance, we employ multiple standard classification metrics:

Accuracy: Measures the overall correctness of the classification model, representing the proportion of correctly predicted instances among all predictions.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

Where TP (True Positives) represents correctly predicted positive instances, TN (True Negatives) represents correctly predicted negative instances, FP (False Positives) represents incorrectly predicted positive instances, and FN (False Negatives) represents incorrectly predicted negative instances.

Precision: Measures the reliability of positive predictions, indicating the proportion of correctly classified positive instances among all instances predicted as positive.

$$\text{Precision} = \frac{TP}{TP + FP}$$

High precision indicates that when the system predicts a requirement belongs to a category, it is likely correct.

Recall (Sensitivity): Measures the ability of the model to identify all relevant positive instances, representing the proportion of actual positive cases correctly detected.

$$\text{Recall} = \frac{TP}{TP + FN}$$

High recall indicates that the system successfully identifies most requirements belonging to each category.

F1-Score: The harmonic mean of precision and recall, providing a balanced measure that considers both metrics.

$$F1\text{-}Score = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

The F1-score is particularly useful when there is a need to balance precision and recall, providing a single measure that reflects both accuracy of predictions and completeness of detection.

Using multiple metrics ensures a comprehensive assessment of model performance. While accuracy provides an overall measure of correctness, precision and recall help analyze the quality of predictions and the model's ability to detect relevant requirement statements. The F1-score further balances these aspects, offering a more reliable performance indicator.

5) Ambiguity Detection Evaluation

For ambiguity detection, we evaluated the system's ability to identify ambiguous requirements. In addition to the standard classification metrics above, we conducted error analysis to

understand the types of ambiguities detected and missed by the system.

VI. Experimental Results and Evaluation

This section presents experimental results evaluating the proposed AI-powered requirement analysis system.

6) *Classification Performance*

Table III presents the overall performance of the ensemble classification model across all requirement types, based on 10-fold cross-validation.

Metric	Value (%)	Standard Deviation (%)
Accuracy	91.2	2.3
Precision (Macro Avg)	89.4	2.8
Recall (Macro Avg)	87.1	3.1
F1-Score (Macro Avg)	88.2	2.7

The system achieves 91.2% accuracy, indicating strong overall performance in classifying requirement statements. The macro-averaged F1-score of 88.2% demonstrates balanced performance across requirement categories, with relatively small standard deviations indicating stability across cross-validation folds.

7) *Performance by Requirement Type*

Table IV presents classification performance broken down by the requirement type.

Requirement Type	Precision (%)	Recall (%)	F1-Score (%)	Support
Functional	92.3	94.1	93.2	1,452
Non-Functional	87.6	84.2	85.9	895
Security	88.1	83.5	85.7	312
Interface	84.5	79.8	82.1	188
Weighted Avg	**89.8**	**88.4**	**89.1**	**2,847**

Functional requirements achieved the highest F1-score (93.2%), reflecting their prevalence in the dataset and relatively distinctive linguistic patterns. Non-functional and security requirements showed similar performance (F1-scores of 85.9% and 85.7% respectively), while interface requirements proved most challenging (F1-score of 82.1%), likely due to their smaller sample size and greater linguistic variability.

8) *Comparison of Classification Approaches*

To validate our ensemble approach, we compared its performance against individual classifiers on a held-out test set (20% of data). Table V presents this comparison.

Approach	Accuracy (%)	F1-Score (%)	Training Time (s)	Inference Time (ms/req)
Logistic Regression	86.3	84.7	12.4	0.3
Random Forest	88.9	87.2	48.7	0.8
BERT-base (fine-tuned)	92.7	91.5	3,245	45.2
Ensemble (RF+LR)	**91.2**	**89.8**	**61.2**	**1.1**

Ensemble + BERT	93.1	92.3	3,306	46.3

The ensemble approach (Random Forest + Logistic Regression) achieves strong performance (91.2% accuracy, 89.8% F1-score) while maintaining practical training and inference times. The BERT-based model achieves the highest accuracy (92.7%) but at a substantially higher computational cost. The full ensemble including BERT offers marginal improvement (0.9% accuracy gain) at nearly 50× the inference cost per requirement, suggesting that for many applications, the computationally efficient ensemble provides the best balance of performance and practicality.

N. Ambiguity Detection Performance

The ambiguity detection module was evaluated on requirements flagged as ambiguous in the annotated dataset. Table VI presents these results.

Metric	Value (%)
Accuracy	87.3
Precision	84.6
Recall	82.1
F1-Score	83.3

The system achieves 83.3% F1-score in detecting ambiguous requirements. Error analysis revealed that the system performs best on lexical ambiguities (vague terms, subjective language) and struggles more with pragmatic ambiguities that require broader contextual or domain knowledge to identify.

O. Efficiency Analysis

We measured processing efficiency on documents of varying sizes to assess practical applicability.

Table VII presents timing results averaged across 10 runs.

Document size (requirement)	Processing Time (Second)	Time Per Requirement
50	0.8	16.0
200	2.9	14.5
500	7.1	14.2
1,000	14.3	14.3
2,000	29.1	14.6

Processing time scales approximately linearly with document size, with average processing time of 14-16 milliseconds per requirement. A typical SRS document containing 200-500 requirements can be fully analyzed in 3-7 seconds, representing substantial efficiency gains compared to manual analysis requiring hours.

Manual analysis, by contrast, is subject to inter-analyst variability and intra-analyst variability over time. Different analysts may interpret requirements differently, and the same analyst may apply different standards when fatigued or distracted.

P. Comparison with Manual Analysis Baseline

Dimension	AI-Powered System	Manual Analysis (Average)
Accuracy (%)	91.2%	88.7%
Time for 100 requirements	1.5 seconds	47 minutes
Consistency (variance)	0% (deterministic)	5.2% across analysts

VII. DISCUSSION

A. *Interpretation of Findings*

The experimental results demonstrate that the proposed AI-powered requirement analysis system achieves strong performance across classification and ambiguity detection tasks. Several findings warrant discussion:

Performance Benchmarking: Our ensemble approach achieves 91.2% accuracy, which is competitive with state-ofthe-art results reported in the literature (97% in Izhar et al. [3] and >81% in Abdeahad et al. [4]). The difference between our results and Izhar et al. may be attributable to dataset differences, as their study employed a different corpus with potentially more distinctive linguistic patterns.

Computational Efficiency: The ensemble model provides an excellent balance of performance and efficiency. While BERT-based models offer marginal accuracy improvements (approximately 1.5%), they require substantially greater computational resources. For organizations processing large volumes of requirements, the ensemble approach may represent the optimal trade-off.

Ambiguity Detection Challenges: The lower performance on ambiguity detection (83.3% F1) compared to classification (91.2% accuracy) reflects the inherent difficulty of this task. Ambiguity is often subjective and context-dependent, making it challenging even for human analysts. The error analysis suggests that pragmatic ambiguities requiring domain knowledge remain particularly challenging.

Practical Viability: The processing speed of 14-16 milliseconds per requirement indicates that the system can analyze a typical SRS document in seconds, enabling integration into continuous integration pipelines and agile development workflows. This represents a qualitative shift from point-in-time manual analysis to continuous automated validation.

B. Comparison with Literature

Our findings align with and extend the literature synthesized in Section II. The hybrid approach combining multiple ML techniques, as advocated by Izhar et al. [3], proved effective in our ensemble architecture. The importance of dataset quality and diversity, noted across multiple studies [3], [4], [7], was confirmed through our careful dataset construction and annotation.

Our computational efficiency analysis extends the literature by quantifying the practical trade-offs between model sophistication and processing speed. While Benitez-Andrades et al. [9] noted the high computational cost of BERT models, our results provide concrete timing comparisons that organizations can use for capacity planning.

The persistent need for human oversight, emphasized by Habiba et al. [7] and Sapate et al. [8], is reflected in our ambiguity detection performance. The 83.3% F1-score indicates that while the system can significantly reduce manual effort, human review remains necessary, particularly for complex or ambiguous requirements.

C. Limitations

This study has several limitations that should be acknowledged:

Dataset Limitations: While we construct a diverse dataset, it may not fully represent the range of requirements encountered in industrial practice. Proprietary requirements from commercial projects could not be included due to confidentiality constraints.

Domain Coverage: The dataset includes requirements from multiple domains, but domain-specific performance may vary. Organizations in specialized domains (e.g., medical devices, aerospace) should validate system performance on their specific requirement of corpora.

Ambiguity Annotation Subjectivity: Ambiguity annotation inherently involves subjectivity. While we achieved substantial inter-annotator agreement (κ=0.79), some ambiguity judgments remain debatable.

Limited Manual Comparison: Our manual analysis comparison, while informative, involved only three analysts and 100 requirements. Larger-scale comparisons would provide more robust benchmarks.

Language Limitation: The system currently supports only English-language requirements. Multilingual support would be necessary for global development contexts.

D. Implications for Practice

The findings have several implications for software engineering practice:

Integration into Development Workflows: The system's processing speed enables integration into continuous integration pipelines, allowing requirements validation to

occur alongside code validation. This supports agile methodologies where requirements evolve incrementally.

Augmentation, Not Replacement: The results support a view of AI as augmenting human analysts rather than replacing them. The system can handle routine classification and flag potential issues, allowing human experts to focus on complex cases, stakeholder negotiation, and strategic decisions.

Training Data Quality: The importance of high-quality training data underscores the need for organizations to invest in systematic requirements annotation and management. Welldocumented historical requirements provide the foundation for effective AI systems.

Cost-Benefit Considerations: Organizations should consider the trade-offs between accuracy and computational cost when selecting AI approaches. For many applications, computationally efficient ensemble methods may provide sufficient accuracy at a substantially lower cost than deep learning alternatives.

VIII. CONCLUSION AND FUTURE WORK

A. *Summary of Contributions*

This paper has presented a comprehensive investigation of AI-powered software requirement document analysis, with two primary contributions:

Systematic Literature Synthesis: We synthesized findings from ten recent research papers, identifying that AI-powered systems can achieve up to 97% accuracy in ambiguity detection, with hybrid approaches combining rule-based and machine learning techniques proving particularly effective. The synthesis revealed persistent challenges in domain adaptation, computational efficiency, and the need for human oversight.

Proposed System and Evaluation: Based on literature insights, we proposed and evaluated a comprehensive AI-powered requirement analysis system. Experimental results demonstrate that our ensemble approach achieves 91.2% accuracy in requirement classification and 83.3% F1-score in ambiguity detection, with processing speeds of 14-16 milliseconds per requirement—representing substantial efficiency gains over manual analysis.

B. *Key Findings*

The key findings from this research are:

AI-powered systems are highly effective for requirement classification and ambiguity detection, achieving accuracy competitive with or exceeding human analysts for welldefined tasks.

Hybrid and ensemble approaches provide an excellent balance of performance and computational efficiency, making them practical for real-world deployment.

Processing speed enables continuous validation, with typical SRS documents analyzed in seconds rather than hours, supporting agile and iterative development approaches.

Human oversight remains necessary, particularly for ambiguous requirements, domain adaptation, and validation of critical requirements.

Dataset quality is paramount, with diverse, well-annotated training data essential for system effectiveness.

C. *Future Work*

Several directions for future research emerge from this work:

Larger-Scale Industrial Validation: Deploying and evaluating the system in industrial settings would provide insights into real-world performance, integration challenges, and organizational adoption factors.

Enhanced Ambiguity Detection: Developing more sophisticated approaches for pragmatic ambiguity detection, potentially incorporating domain knowledge bases and broader document context, could improve ambiguity detection performance.

Multilingual Support: Extending the system to support multiple languages would enable application in global software development contexts.

Explainable AI Integration: Incorporating explainability techniques would help stakeholders understand and trust system outputs, addressing the interpretability limitation noted in the literature.

Continuous Learning: Developing mechanisms for the system to learn from human feedback over time could improve performance and adaptation to evolving requirements practices.

Integration with Requirements Management Tools: Seamless integration with popular requirements management platforms would facilitate adoption and workflow integration.

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APPENDIX: GLOSSARY OF TERMS

Accuracy: The proportion of correctly classified instances among all instances.

Ambiguity: The property of having multiple possible interpretations, leading to uncertainty in meaning.

BERT (Bidirectional Encoder Representations from Transformers): A transformer-based machine learning technique for natural language processing pre-training.

Bi-LSTM (Bidirectional Long Short-Term Memory): A recurrent neural network architecture that processes sequences in both forward and backward directions.

F1-Score: The harmonic mean of precision and recall, providing a balanced measure of model performance.

Functional Requirement: A statement describing a specific system behavior, function, or feature.

Lemmatization: The process of reducing words to their base or dictionary form.

Non-Functional Requirement: A statement specifying quality attributes, constraints, or performance characteristics.

Precision: The proportion of correctly classified positive instances among all instances predicted as positive.

Recall: The proportion of actual positive instances correctly identified by the model.

SRS (Software Requirements Specification): A document that describes the intended behavior, features, and constraints of a software system.

TF-IDF (Term Frequency-Inverse Document Frequency): A numerical statistic reflecting the importance of a word in a document relative to a collection of documents.

Tokenization: The process of dividing text into individual units (tokens) such as words or punctuation.